

DIGITAL IMAGE PROCESSING

LECTURE # 7

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-III

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2

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Topics to Cover

- Sharpening Spatial Filtering
- Edge Detection
- Derivatives
 - ❑ 1st Order Derivative
 - ❑ 2nd Order Derivative
- Laplacian Operator
- Unsharp Masking
- High Boost Filtering
- Gradient Operators
 - ❑ **Sobel Operator**
 - ❑ **Prewitt Operator**
 - ❑ **Robert Cross Operator**

Spatial Filtering for Sharpening

4

Background: To highlight fine detail in an image or to enhance blurred detail

Applications: Medical imaging, industrial inspection etc.

Foundation (Blurring vs Sharpening):

- **Blurring/smoothing** is performed by spatial averaging (equivalent to integration)
- **Sharpening** is performed by noting only the gray level changes in the image that is the differentiation



Edge Detection

5

What is an Edge?

Edge is a change but every change is not an edge

Edge is a noticeable or abrupt change

E.g.

5	7
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 2 is not a noticeable change in the range of (0 to 255)

We have to define a threshold if the change is more than a specified threshold then we will define it as an edge point.

Here gradual change exists you cannot pinpoint where the edge exists so the change must be abrupt

50	55	60	65	70
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For each pixel we have to look in horizontal, vertical and diagonal direction

$dx/ds \rightarrow$ for horizontal direction

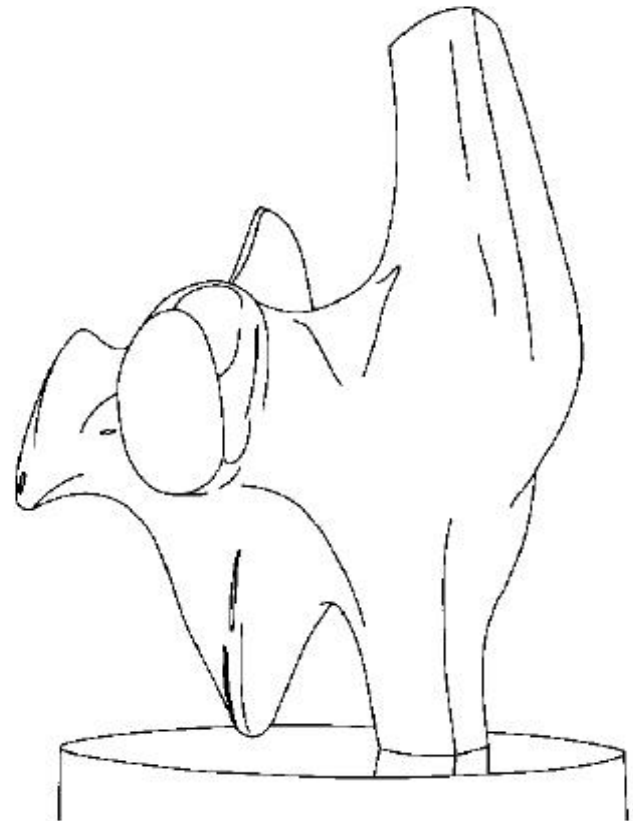
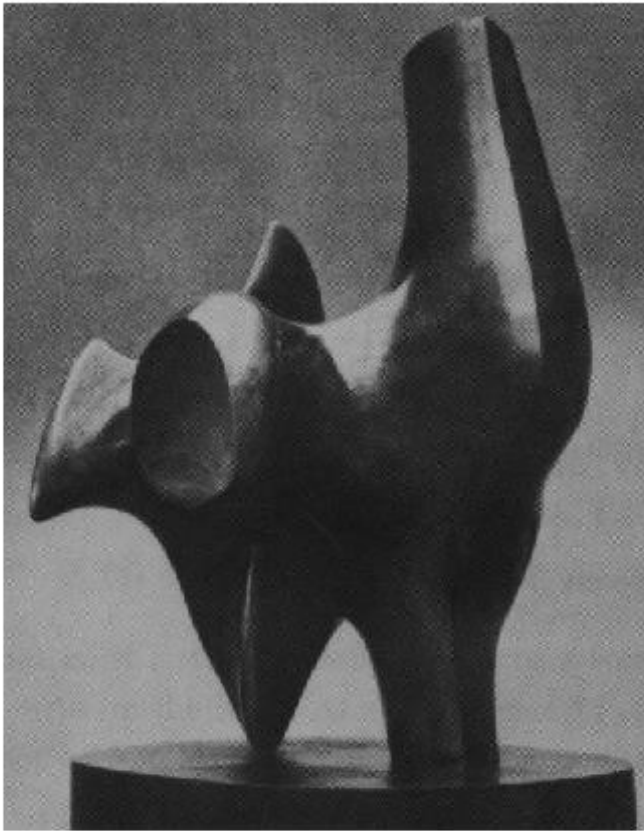
$dy/ds \rightarrow$ for vertical direction

$dd/ds \rightarrow$ for diagonal direction

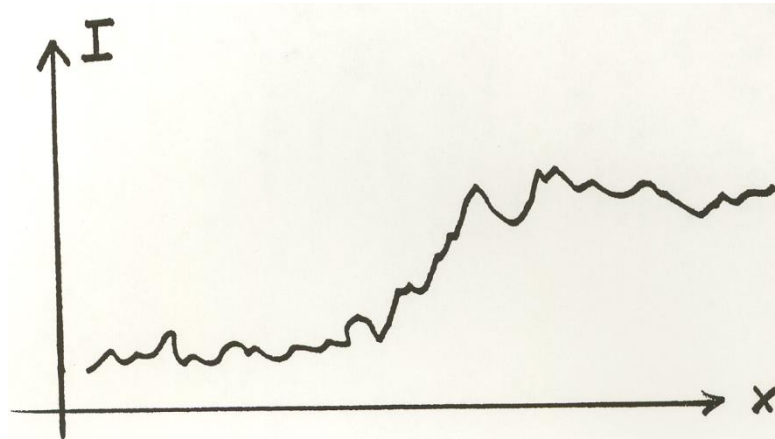
50	15	35
40	20	30
225	195	230

Edge Detection

6



Real Edges



Noisy and Discrete!

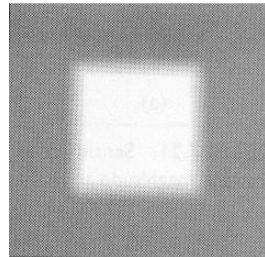
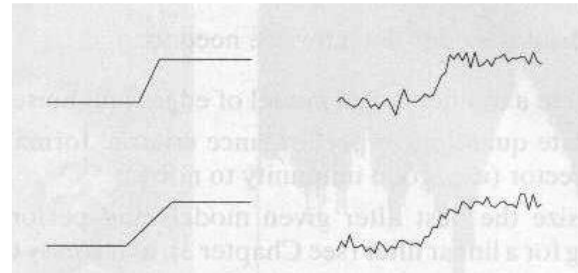
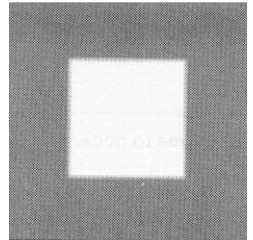
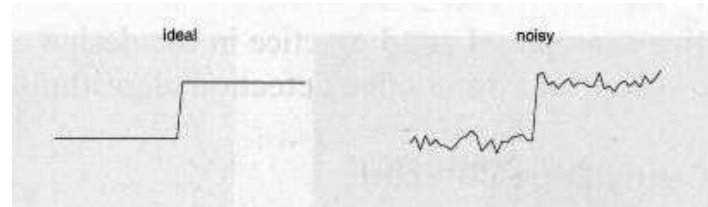
We want an **Edge Operator** that produces:

- ▣ Edge Magnitude
- ▣ Edge Orientation
- ▣ High Detection Rate and Good Localization

Edge Types

8

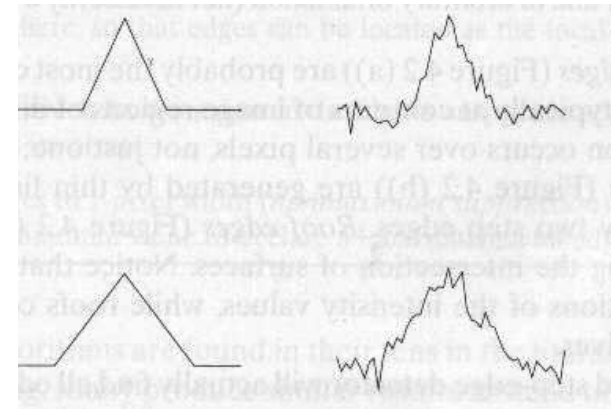
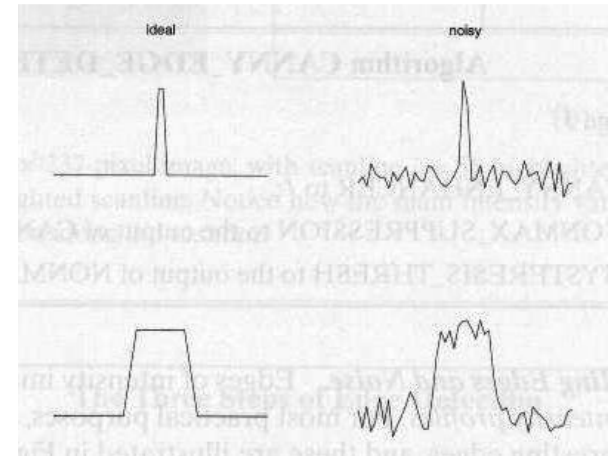
- ❑ **Step edge:** the image intensity abruptly changes from one value on one side of the discontinuity to a different value on the opposite side.
- ❑ **Ramp edge:** a step edge where the intensity change is not instantaneous but occurs over a finite distance.



Edge Types

9

- ❑ Ridge edge: the image intensity abruptly changes value but then returns to the starting value within some short distance (i.e., usually generated by lines).
- ❑ Roof edge: a ridge edge where the intensity change is not instantaneous but occur over a finite distance.



Spatial Filtering for Sharpening

10

Operation of Image Differentiation

- ❑ Enhance edges and discontinuities (magnitude of output gray level $> T$)
- ❑ De-emphasize areas with slowly varying gray-level values (output gray level: 0)

Mathematical Basis of Filtering for Image Sharpening

- ❑ First-order derivatives [Gradient]
- ❑ Second-order derivatives

Derivatives

11

First Order Derivative

- A basic definition of the first-order derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

Second Order Derivative

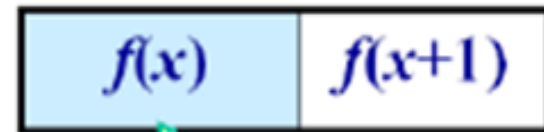
- Similarly, we define the second-order derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

1st & 2nd Order Derivatives

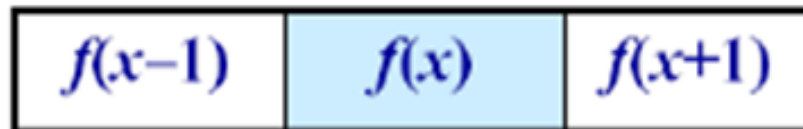
12

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$



Position for the
output pixel

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



1st & 2nd Order Derivatives

13

First Order Derivative

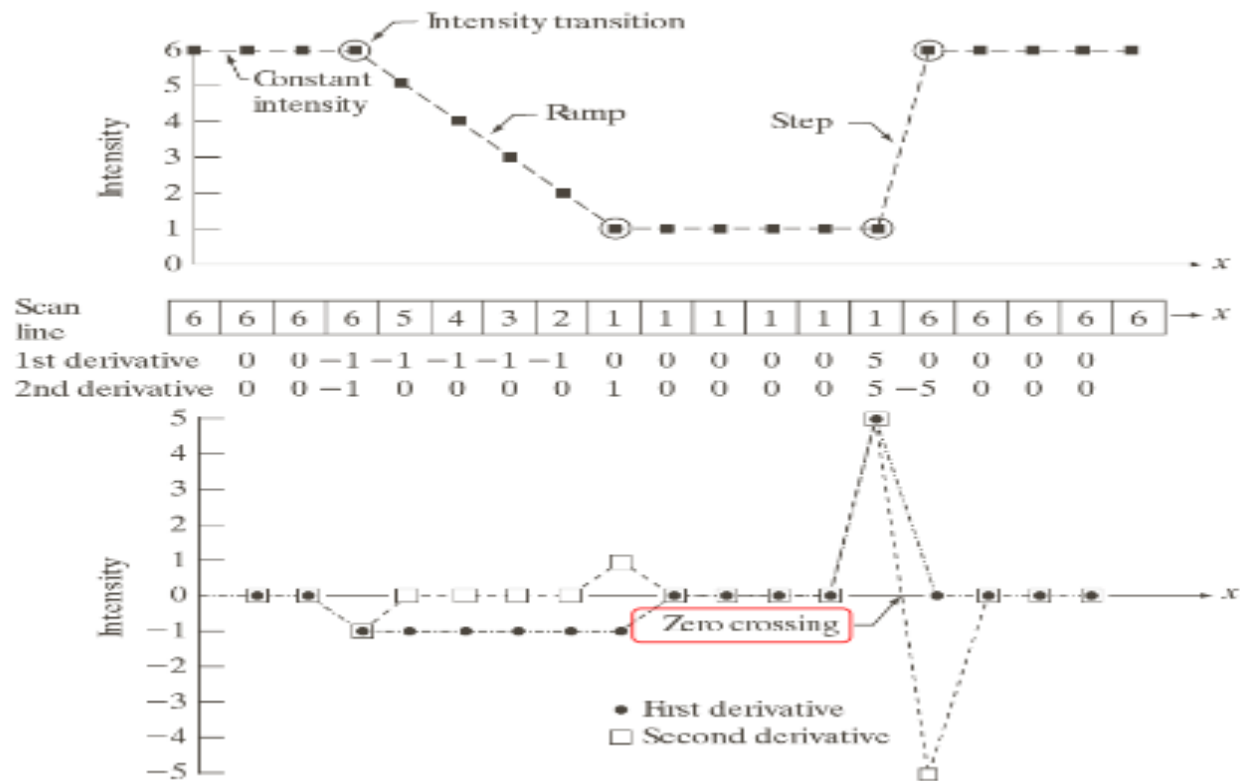
- ❑ Must be zero in area of constant gray levels
- ❑ Non zero along the ramps
- ❑ Non zero at the start of the gray level step or ramp

Second Order Derivative

- ❑ Zero in flat areas
- ❑ Zero along the ramps of constant slope
- ❑ Non zero at the start and end of the gray level step or ramp

Example for Discrete Derivatives

14

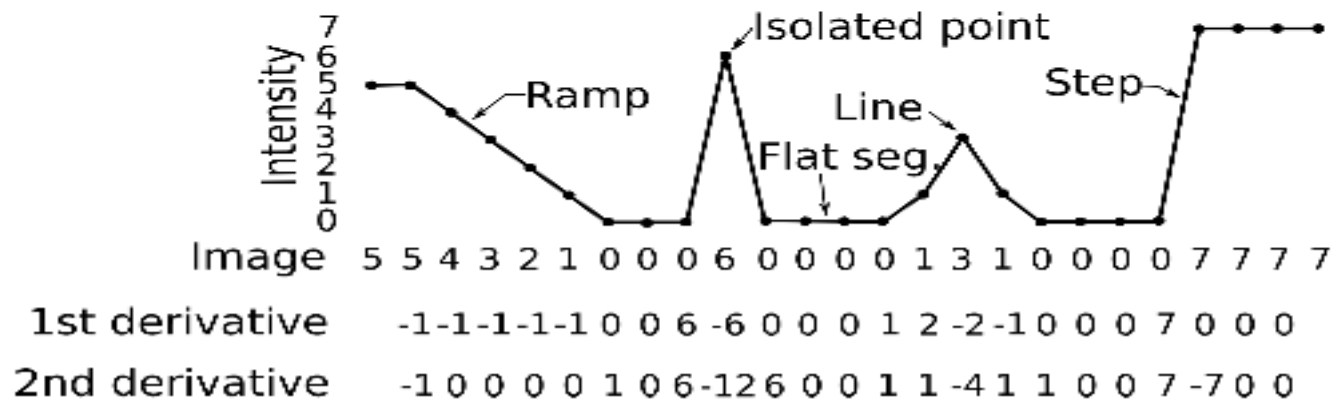
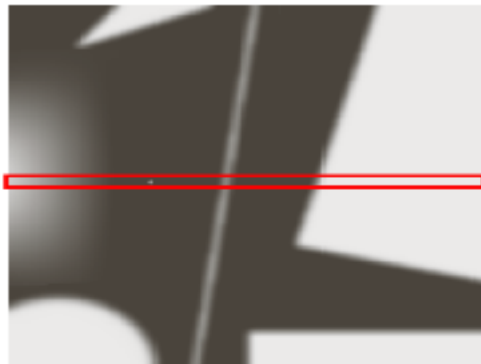


a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

Example for Discrete Derivatives

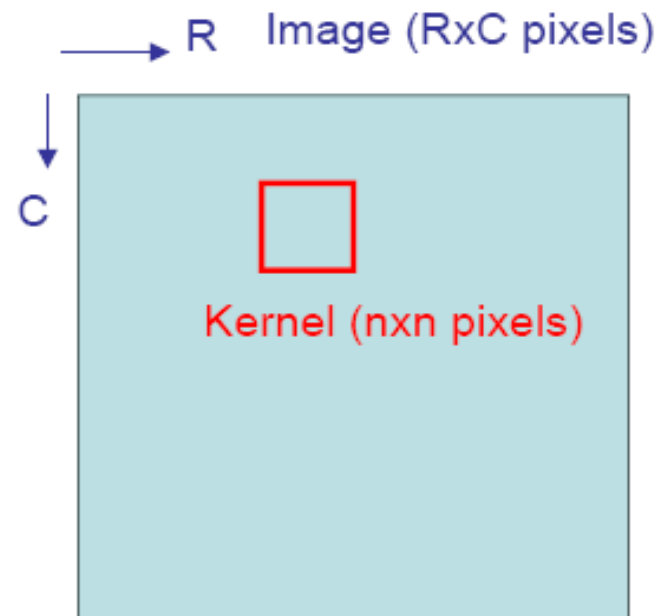
15



Steps to Apply Edge Detector Kernel

16

1. Suppose a $n \times n$ (n is an odd number) kernel is used and the centre position of the kernel is denoted by kc .
2. Slide the kernel across the image, one pixel at a time.
3. Compute the new value of each pixel, which is under kc , and repeat for all pixels in the image.
4. Note that the process is not in-place (new values are stored in a new image of same dimension).
5. Computation:
 - For each pixel inside the kernel, multiply the pixel value with the kernel value.
 - Take the sum of the products.
6. (a) The border pixel can be duplicated, or (b) take the mirror values of the border pixels, to fill up the kernel.



1st Derivative Filtering- The Gradient

17

- 1st derivative filters is Gradient which represents change
- For a function $f(x,y)$ the gradient of f at coordinates (x,y) is given as the column vector:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

1st Derivative Filtering- The Gradient

18

- The magnitude of this vector is given by:

$$\begin{aligned}\nabla f &= \text{mag}(\nabla f) \\ &= [G_x^2 + G_y^2]^{1/2} \\ &= \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}\end{aligned}$$

- The direction of this vector is given by:

$$\theta = \tan^{-1} [G_y / G_x]$$

1st Derivative Filtering- The Gradient

19

- Now we want to define **digital approximations** and their Filter Masks
- For simplicity we use a 3x3 region
- For example, **z5** denotes $f(x,y)$, **z1** denotes $f(x-1,y-1)$
- A simple approximation for First Derivative is

$$G_x = (z_8 - z_4) \text{ and } G_y = (z_6 - z_2)$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

1st Derivative Filtering- The Gradient

20

- Two other definitions proposed by Roberts use cross- difference

$$G_x = (z_9 - z_5) \quad \text{and} \quad G_y = (z_8 - z_6).$$

- If we use

$$\nabla f = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

$$\nabla f = [(z_9 - z_5)^2 + (z_8 - z_6)^2]^{1/2}$$

-1	0
0	1

0	-1
1	0

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Roberts Cross-Gradient Operators

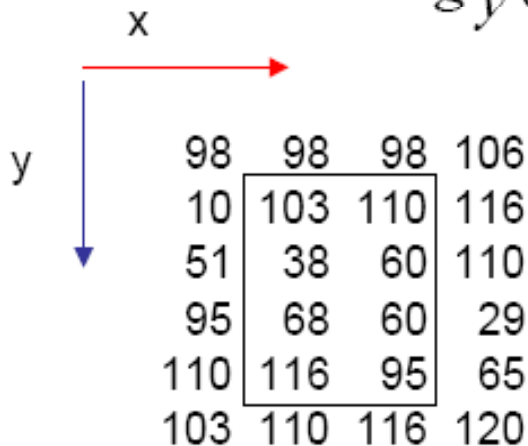
A Simple Edge Detector- The Gradient

21

- Based on *grey scale gradient* at a pixel

$$g_x(x, y) \approx f(x+1, y) - f(x-1, y)$$

$$g_y(x, y) \approx f(x, y+1) - f(x, y-1)$$



A Simple Edge Detector- The Gradient

22

- Based on *grey scale gradient* at a pixel

$$g_x(x, y) \approx f(x+1, y) - f(x-1, y)$$

$$g_y(x, y) \approx f(x, y+1) - f(x, y-1)$$

x →

y ↓

98	98	98	106
10	103	110	116
51	38	60	110
95	68	60	29
110	116	95	65
103	110	116	120

→

100	13	-60	-38
9	72	-35	-50
-35	39	78	35
-15	51	42	56

↓

What is

$$\sqrt{g_x^2 + g_y^2} ?$$

Answer:

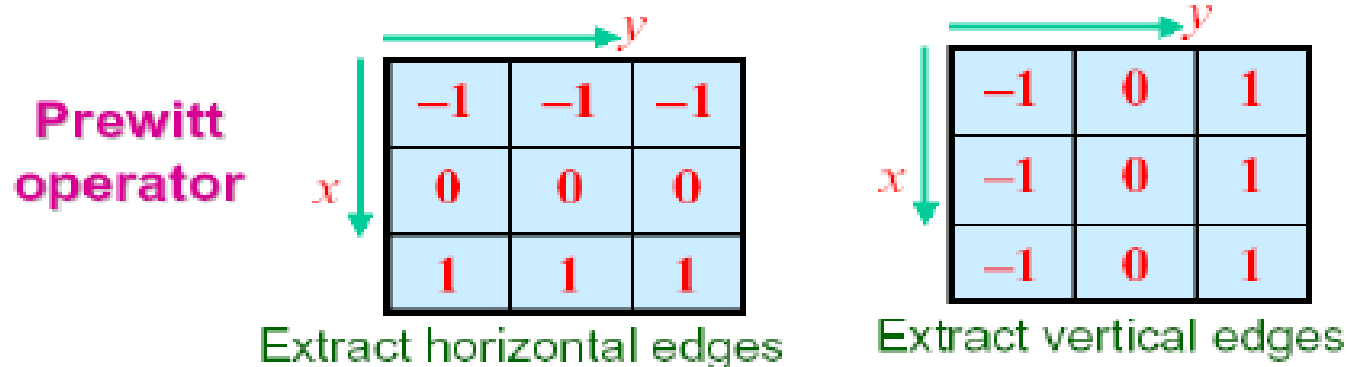
117	40
36	87
85	52
44	76

Assuming threshold $T = 50$,
a pixel is selected if $\geq T$

Gradient Operators

23

- Normally the smallest mask used is of size 3 x 3
- Based on the concept of approximating the gradient several spatial masks have been proposed



Pixel
arrangement

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Prewitt operators (equation):

$$\nabla f \approx \left| (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3) \right| + \left| (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7) \right|$$

Gradient Operators

24

$$h_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad h_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Assuming threshold = 50

(1) What are the detected pixels?

98	98	98	106
10	103	110	116
51	38	60	110
95	68	60	29
110	116	95	65
103	110	116	120

→
109 93
74 46
-41 -18
-37 -80

↓
-145 -94
0 -172
172 68
106 189

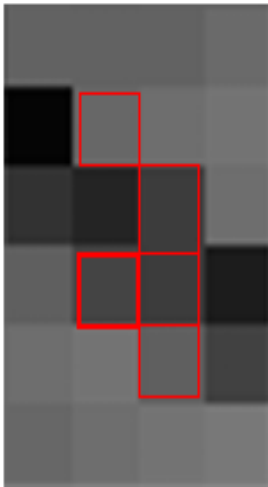
(2) What will happen if the threshold is 100? 150?

Gradient Operators

25

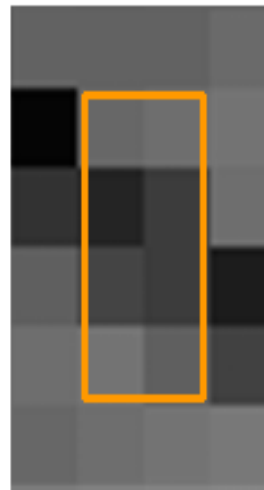
One edge but thick

Result from gradient



Result from Prewitt kernels

Threshold=50

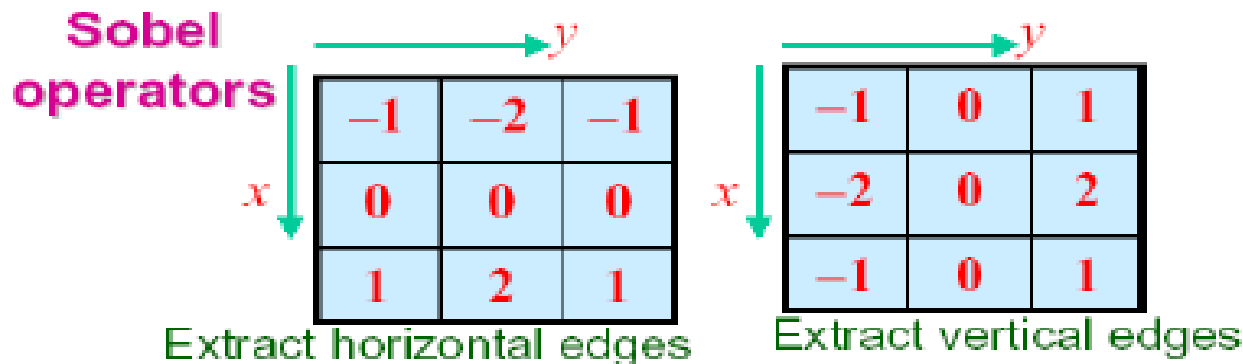


Threshold=100 or 150



Gradient Operators

26



Sobel operators (equation):

Emphasize more the current position (y)

Pixel arrangement

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$\nabla f \approx \left| (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \right| + \left| (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \right|$$

Emphasize more the current position (x)

Sharpening Mask Coefficients

27

- Why the summation of coefficients in all masks of derivate operators equals to zero?

Laplacian Mask: 2nd Order Derivative

28

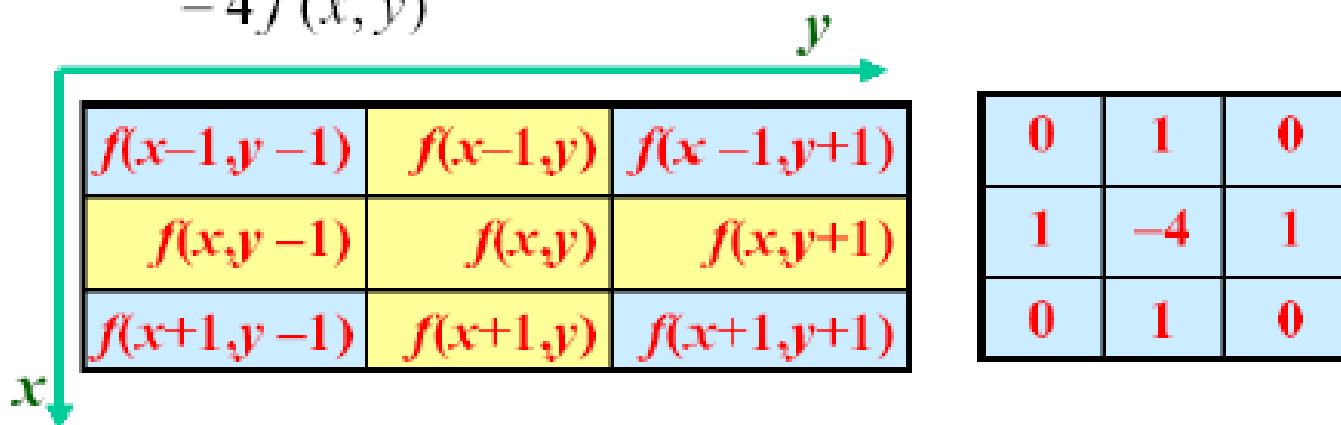
Laplacian operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



Laplacian Mask: 2nd Order Derivative

29

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a	b
c	d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).
(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian for Image Enhancement

30

- Image background is removed by Laplacian filtering.
- Background can be recovered simply by adding original image to Laplacian output

To obtain the enhanced image $\hat{p}$

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y), w_5 < 0 \\ f(x, y) + \nabla^2 f(x, y), w_5 > 0 \end{cases}$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

In this way, background tonality can be perfectly preserved while details are enhanced

Laplacian for Image Enhancement

31

a	b
c	d

FIGURE 3.46

(a) Blurred image of the North Pole of the moon.

(b) Laplacian image obtained using the kernel in Fig. 3.45(a).

(c) Image sharpened using Eq. (3-54) with $c = -1$.

(d) Image sharpened using the same procedure, but with the kernel in Fig. 3.45(b).

(Original image courtesy of NASA.)

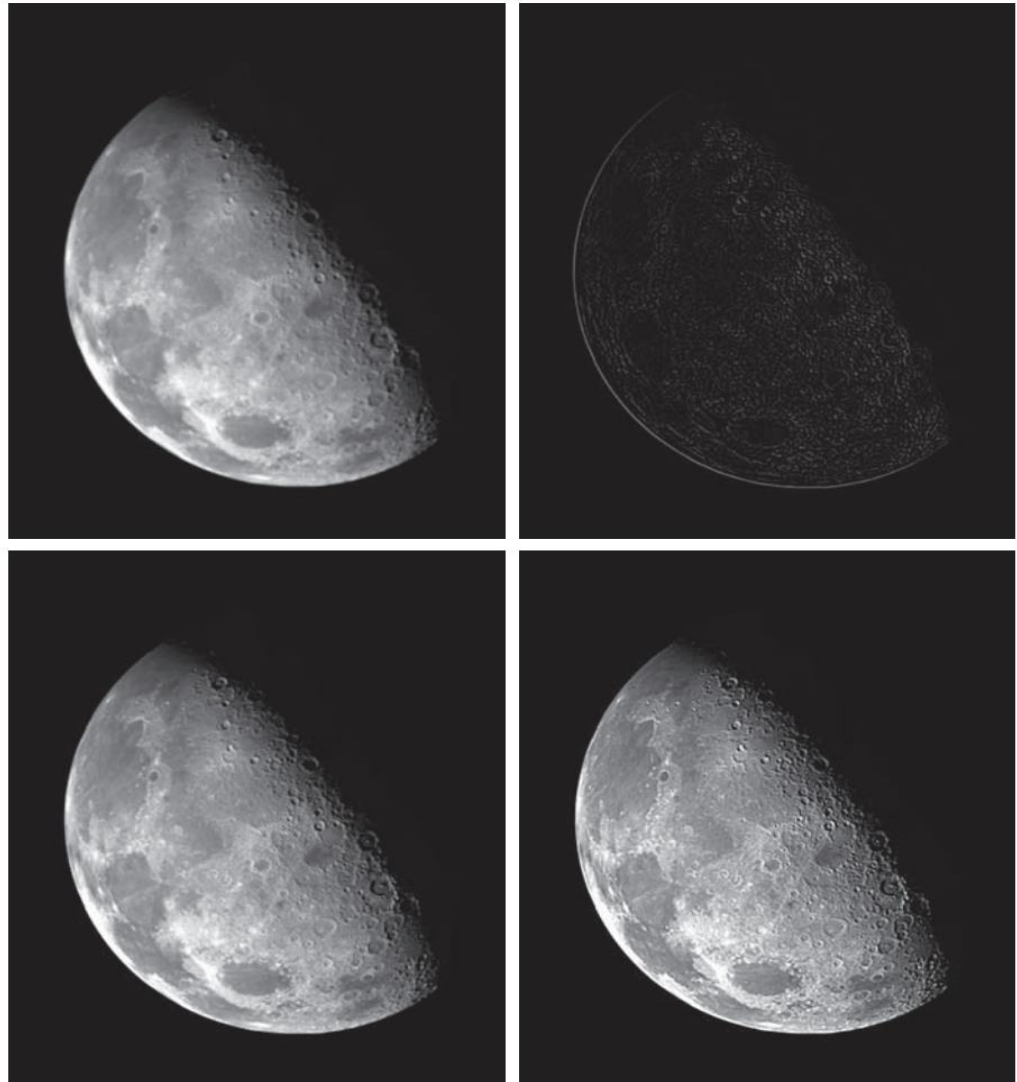


Image Sharpening Based on Un-sharp Masking

32

Un-sharp masking

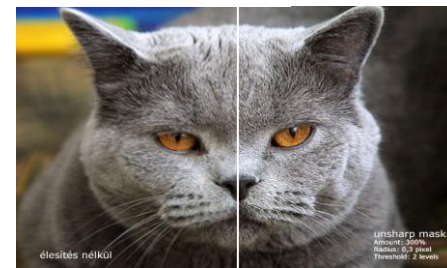
- Subtracting an unsharp (smoothed) version of an image from the original image is process that has been used since the 1930s by the printing and publishing industry to sharpen images. This process, called unsharp masking, consists of the following steps:
 1. Blur the original image.
 2. Subtract the blurred image from the original (the resulting difference is called the mask.)
 3. Add the mask to the original.

Letting $\bar{f}(x, y)$ denote the blurred image, the mask in equation form is given by:

$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y) \quad (3-55)$$

Then we add a weighted portion of the mask back to the original image:

$$g(x, y) = f(x, y) + k g_{\text{mask}}(x, y) \quad (3-56)$$



High Boost Filtering

33

- A slight further generalization of un-sharp masking is called High Boost filtering
- A high boost filtered image, f_{hb} , is defined at any point (x, y) as

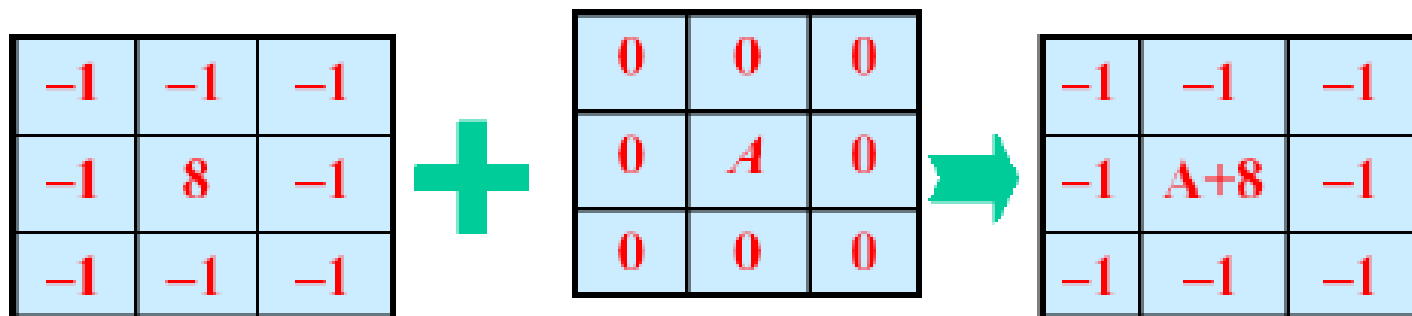
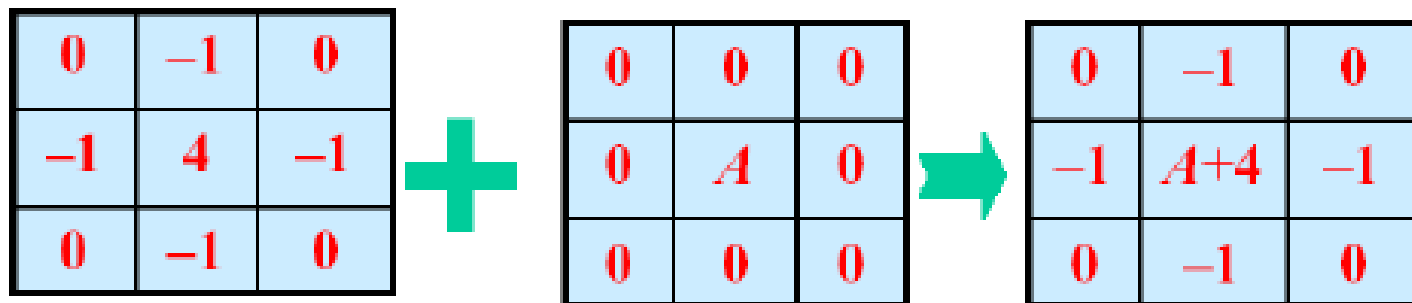
$$\begin{aligned} f_{hb}(x, y) &= Af(x, y) - \bar{f}(x, y), A > 1 \\ &= (A-1)f(x, y) + f(x, y) - \bar{f}(x, y) \\ &= (A-1)f(x, y) + f_s(x, y) \\ &\xrightarrow[\text{to } f_s(x, y)]{\text{apply Laplacian mask}} = (A-1)f(x, y) + [f(x, y) \pm \nabla^2 f(x, y)] \\ &= Af(x, y) \pm \nabla^2 f(x, y) \end{aligned}$$

Principal application:

- High Boost filtering is used when input image is darker than desired
- High-boost filter makes the image lighter and more natural

High Boost Filtering

34



$A \geq 1$

High Boost Filtering Masks

35

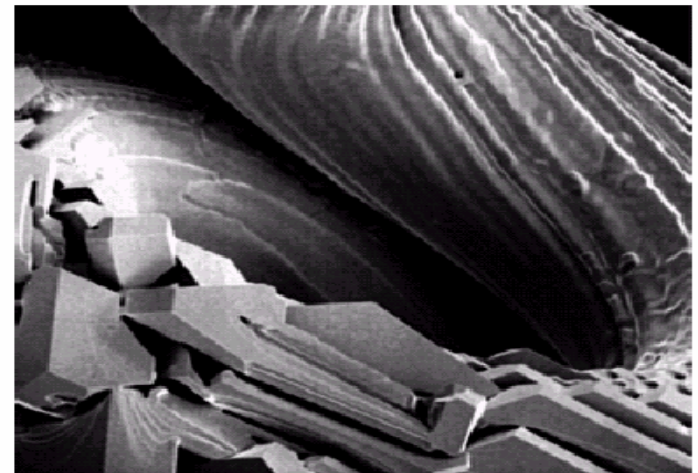
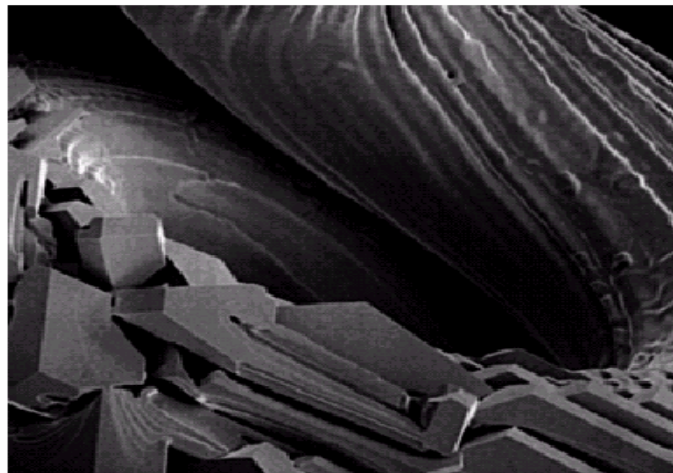
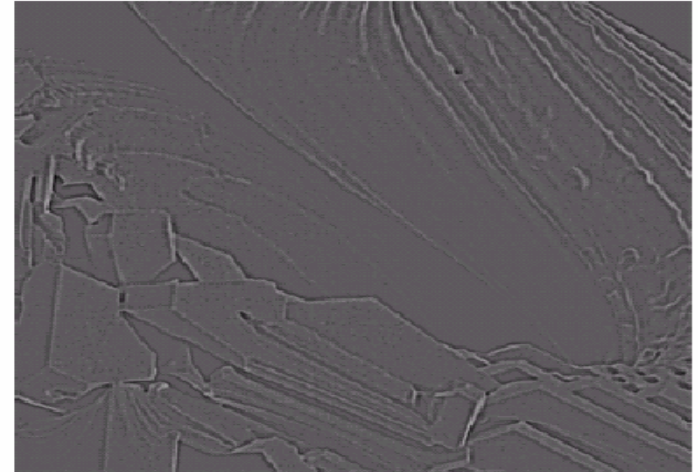
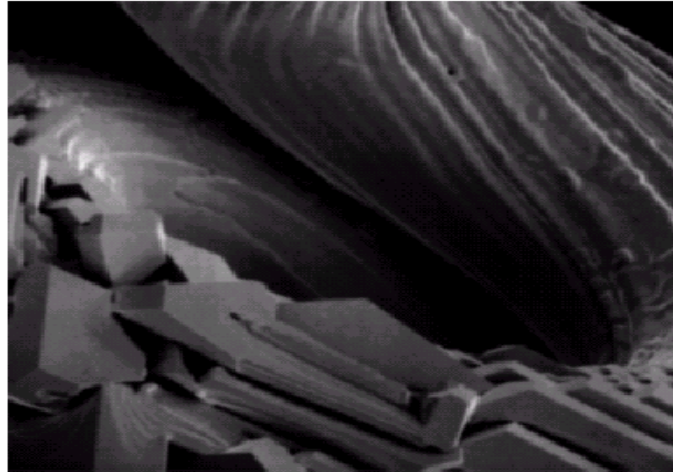
a	b
c	d

FIGURE 3.43

(a) Same as Fig. 3.41(c), but darker.

(b) Laplacian of (a) computed with the mask in Fig. 3.42(b) using $A = 0$.

(c) Laplacian enhanced image using the mask in Fig. 3.42(b) with $A = 1$. (d) Same as (c), but using $A = 1.7$.



High Boost Filtering Masks

36

Original Image



0	-1	0	-1	-1	-1
-1	$A + 4$	-1	-1	$A + 8$	-1
0	-1	0	-1	-1	-1

High Boost Filtering Masks

37



Mask 1, $A = 1$



Mask 1, $A = 1.5$



Mask 1, $A = 2$



Mask 2, $A = 1$



Mask 2, $A = 1.5$



Mask 2, $A = 2$

References

38

1. DIP by Gonzalez

For any query Feel Free to ask



39

